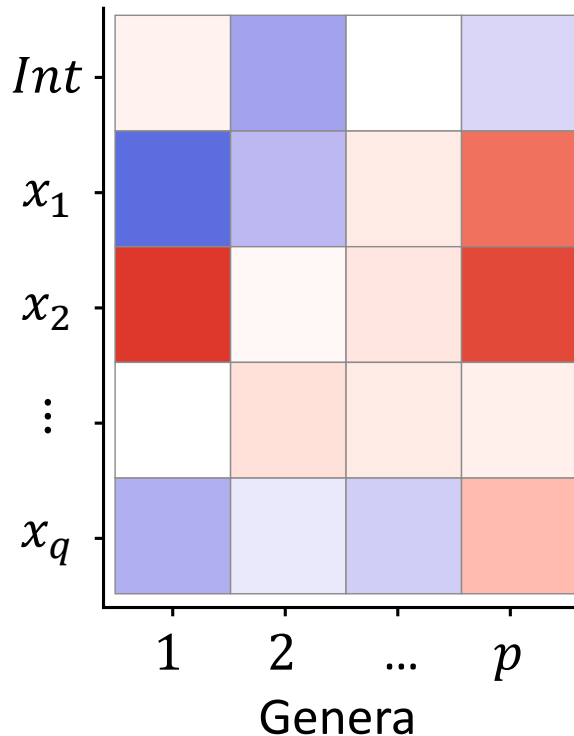


CLR transformed matrix

		Genera				
		1	...	i	...	p
Samples	1	5.139	...	1.393	...	2.374
	$\vdots$	$\vdots$	...	$\vdots$	...	$\vdots$
	k	-1.395	...	3.128	...	1.806
	$\vdots$	$\vdots$	...	$\vdots$	...	$\vdots$
	n	3.841	...	1.896	...	2.301

Multivariate
regression

Coefficient matrix
(feature-level)Test covariate effects
(community-level)

Test the global effect
of covariate h :

$$H_0: \boldsymbol{\beta}_h = \mathbf{0}$$

$$H_1: \boldsymbol{\beta}_h \neq \mathbf{0}$$

($\boldsymbol{\beta}_h$ is the coefficient
vector across genera)